Reading Data into Spark

In this train, we'll explore how to read various types of data into Apache Spark, ensuring efficient data loading for further processing.

Learning Objectives

By the end of this train, you should be able to:

- Read data into Spark from multiple sources.

Load Dependencies

To start, let's set up Spark and read in the Bitcoin dataset.

The dataset can be accessed by downloading the accompanying zip folder. The file to access is: `spark_deeper_concepts_transformed_data.snappy.parquet`

```
import pyspark
from pyspark import SparkContext
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql.types import StringType, StructType, StructField, IntegerType,
FloatType
```

```
sc = SparkContext.getOrCreate()
spark = SparkSession(sc)
```

For the below cell, make sure that you download and unzip the required file and place it in your current working directory.

```
bitcoin_data =
spark.read.parquet('./spark_deeper_concepts_transformed_data.snappy.parquet')
```

```
bitcoin_data.printSchema()
```

Output:

```
root
 |-- timestamp: timestamp (nullable = true)
 |-- open: float (nullable = true)
 |-- high: float (nullable = true)
 |-- low: float (nullable = true)
 |-- close: float (nullable = true)
 |-- volume_btc: float (nullable = true)
 |-- volume_currency: float (nullable = true)
 |-- weighted_price: float (nullable = true)
```

```
bitcoin_data.show(10, False)
```

Output:

```
+-------------------+----+----+----+-----+----------+---------------+--------------+
|timestamp          |open|high|low |close|volume_btc|volume_currency|weighted_price|
+-------------------+----+----+----+-----+----------+---------------+--------------+
|2011-12-31 09:52:00|4.39|4.39|4.39|4.39 |0.45558086|2.0            |4.39          |
|2011-12-31 09:53:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:54:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:55:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:56:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:57:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:58:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 09:59:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 10:00:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
|2011-12-31 10:01:00|NaN |NaN |NaN |NaN  |NaN       |NaN            |NaN           |
+-------------------+----+----+----+-----+----------+---------------+--------------+
only showing top 10 rows
```

Conclusion

In this lesson, you learned how to load data into Spark using the Parquet file format. You also explored how to inspect your data using .printSchema() and .show() to prepare it for further analysis.

▼Key Takeaways ✖

- **Takeaway 1:** Spark can read data efficiently from Parquet files using the `spark.read.parquet()` method.
- **Takeaway 2:** You must unzip and place the dataset in the correct working directory before loading it into Spark.
- **Takeaway 3:** The `.printSchema()` function allows you to inspect the structure and data types of your DataFrame.
- **Takeaway 4:** Use `.show()` to preview the contents of your DataFrame and verify that the data loaded correctly.
- **Takeaway 5:** Before working with data in Spark, ensure SparkContext and SparkSession are correctly initialized.